

Assignment 1

Recreating Figure 1 in Blender Using Three Methods

You are required to recreate Figure 1 in Blender using three different approaches. Each part focuses on a distinct method of modeling.

Part 1: Geometry Nodes (Procedural Method)

Task: Use the Geometry Nodes Editor to create a procedural setup.

Requirement:

- Design a “Pillar Altar” effect similar to Figure 1.
- Use the Distribute Points on Faces node to scatter instances across a surface.
Control Feature:
- Add a Density parameter as an input in your node group.
- This parameter should allow users to control the number of instances directly from the Modifier Properties panel, without opening the node editor.
- If any other node is needed for the environment, you can also add that node.

Part 2: Scripting (Python Method)

Task: Recreate Figure 1 using Blender’s Scripting Workspace.

Requirement:

Use Python (bpy module) to procedurally generate the same scene or effect as in Figure 1.

Part 3: Manual Modeling (Traditional Method)

Task: Recreate Figure 1 using standard modeling tools in Blender.

Requirement:

Use manual techniques such as Extrude, Bevel, Subdivision, and other relevant tools.

Build the model step-by-step without procedural or scripting assistance.



Figure 1(Ignore the texture for now, just apply the material)

Submission Requirements

You must prepare and submit a **.docx (Word document)** containing the following:

1. Part 1: Geometry Nodes

- Insert a **screenshot of your final Geometry Nodes setup**
- A **screenshot of the final output in the 3D viewport.**

2. Part 2: Scripting (Python in Blender)

- Insert:
 - A **screenshot of the final output in the 3D viewport**
 - A **screenshot of the Python script** used in Blender's Scripting Workspace
- Also include the **full Python code** used to generate Figure 1.

3. Part 3: Manual Modeling

- Insert **step-by-step screenshots** showing your workflow in Blender.
- Each step should clearly show progress, such as:
 - Base mesh creation
 - Extrusion / shaping
 - Beveling / subdivision
 - Final scene setup
- The final screenshot of the completed Figure 1 must also be included.

4. Also submit the blender file.